

NEBULA

RL-assisted automated analog circuit design with SKY130 and ngspice

Specifications in. A circuit, measured specifications and an auditable decision out.

Nebula connects a trained policy, deterministic verification, transistor simulation and circuit export in a Python design framework. The policy explores a characterized 512-setting library; safety recovery and target centering make its role inspectable. A separate physical development implements voltage-configurable source degeneration and a transistor one-tap DFE.

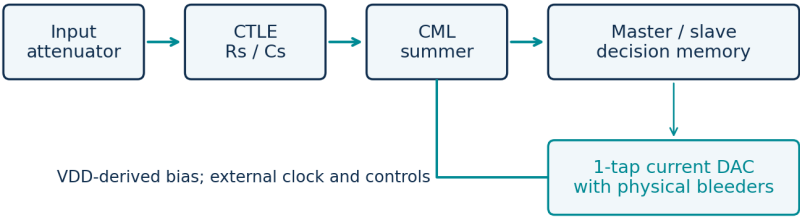


Figure 1. Calibrated transistor checkpoint. The feedback path is physical; clock and control generators remain external.

AUTOMATED FRAMEWORK	CALIBRATED HARDWARE	REVIEWABLE OUTPUT
Target input, policy proposals, verification and exact export	45/45 sampled link PVT; one fixed calibration	Circuit drawings, raw traces, specifications and source hashes

Independent nominal measurements give 8.744-8.747 dB peaking near 1.9 GHz, 0.65405 mVrms held-state noise and approximately -54.92 dBc clocked HD3. The minimum sampled link-PVT eye is 113.244 mV. These are simulation results at declared test conditions, not complete receiver signoff. [E1-E2]

Team

College

Member name	Contact number	Email ID

Contribution. An integrated design workflow, measured policy comparisons and a physical receiver checkpoint make the result inspectable. This report includes all front matter and references within 12 pages.

Requirements and evidence coverage

The requested outcome is an automated design flow whose circuit claims survive verification.

At 5 Gbps, one unit interval is 200 ps and Nyquist is 2.5 GHz. Channel loss spreads each symbol into its neighbors. The prescribed equalizer combines one source-degenerated CTLE stage with variable Rs/Cs and one DFE tap. The official deliverable is an RL-based Python framework: target specifications in, seamless SPICE integration, final schematic and resulting specifications out. LLM-based interaction is a bonus. [E7]

SPECIFICATION	REQUIRED TARGET	EVIDENCE / ASSESSMENT
S1 / Signaling	5 Gbps NRZ, PCIe Gen2	Simulated 200 ps UI; full PCIe protocol compliance is not assessed.
S2 / Topology	CTLE, variable Rs/Cs, one-tap DFE	Implemented in the separate calibrated transistor deck.
S3 / Response	3-12 dB; peak 1.25-2.5 GHz	Loaded nominal 8.744-8.747 dB at 1.903-1.906 GHz. Full target rectangle incomplete.
S4 / Linearity	HD3 below -30 dBc, 100 MHz input	Nominal clocked model HD3 about -54.92 dBc; 100 mV differential peak.
S5 / Noise	Below 1.5 mVrms, 10 MHz-5 GHz	Nominal held-clock noise at most 0.65405 mVrms; periodic noise unverified.
S6 / Power	Below 15 mW	Maximum link-PVT VDD draw 12.5241 mW; external generators excluded.
S7 / Area	Below 0.05 mm ² , 130 nm	0.014715766 mm ² geometry subtotal; not routed layout area.
S8 / Eye	Above 100 mV / 0.4 UI	45-point minima: 113.244 mV, 0.635 UI positive opening; 0.560 UI above 100 mV.
S9 / PVT	5 process x 3 supplies x 3 temperatures	Fixed-control link PVT: 45/45. Simultaneous S3-S8 analog PVT incomplete.

JUDGING CRITERION	WHERE THE SUBMISSION PROVIDES EVIDENCE
Deliverable coverage	Framework/input/output: pp.3,10-11; circuit/specifications: pp.5-9; source guide: p.12.
Innovation	Learned search with explicit verification/recovery, device-to-eye bridge and inspectable provenance: pp.3-4,7,10.
Thought process	Measured failure mechanisms, physical redesign and independent re-verification: pp.6,8-9.

From specification to circuit

A common target contract links user input, learned proposals, verification and output artifacts.

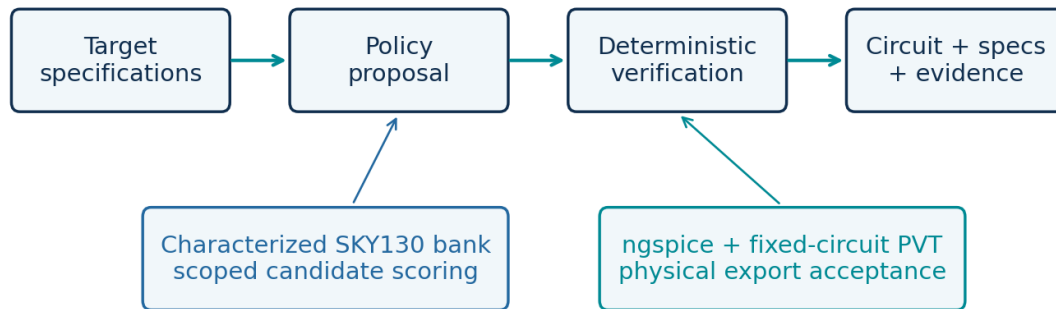


Figure 2. Cached bank scoring and fresh physical acceptance are separate execution paths. The verifier controls acceptance; the policy proposes candidates.

Three evidence layers, three different meanings

LAYER	WHAT RUNS / WHAT IS EXPORTED	WHAT ITS PASS MEANS
Frozen adaptive bank	Policy A/R/C moves; best compliant visit; classical recovery and centering; representative deck plus code map.	Codes may differ across PVT/channel conditions. One representative deck is not the whole map.
Fixed physical CTLE	Eligible fixed setting, physical bias/MIM and exact captured transistor deck. Saved 3, 6 and 9 dB targets at 1.9 GHz.	One fixed CTLE across 315 modeled conditions. Its DFE remains ideal behavioral cancellation.
Integrated checkpoint	73-device configurable CTLE, CML summer, clocked memory and feedback DAC; separate nominal and link-PVT decks.	One loaded 9 dB / 1.9 GHz calibration, independently checked. It was not autonomously discovered by PPO.

Acceptance is a measured engineering decision

The device runner checks operating points, output completeness, model warnings, units and numerical consistency. It reads primitives instead of treating an exit code as proof. The link stage preserves absolute voltage and rejects invalid fits or compression. Each result keeps its request, circuit identity, measured outcomes and exact artifact paths together.

Automatic operation. For a supported request, proposal, selection, verification and export require no manual candidate choice. Topology development, allowed ranges and the later receiver calibration were engineering decisions made before the run. [E1-E3,E6]

What the learned policy does

The learned contribution is a bounded search trajectory through characterized circuits.

PART OF THE MDP	IMPLEMENTED DEFINITION
Observation	62 fields: requested boost/log frequency, trial fraction, current A/R/C codes and measured eye history. PVT, channel loss, compliance, quality and oracle are hidden.
Actions / horizon	A, R or C one index up/down, or LOCK; boundary masks; at most eight measurements, including the fixed start and repeats. Bank = $8 \times 8 \times 8 = 512$ settings.
Quality / reward	q is compliant eye area normalized to the best compliant bank eye area for that identity, clipped to [0,1]. Infeasible candidates have q=0. Reward tracks quality change and penalizes a bad final lock.
Training / acceptance	Imitation initialization then PPO; five frozen seeds. Deployment seed is preselected. The shield retains compliant visits; production adds classical fallback and centering.

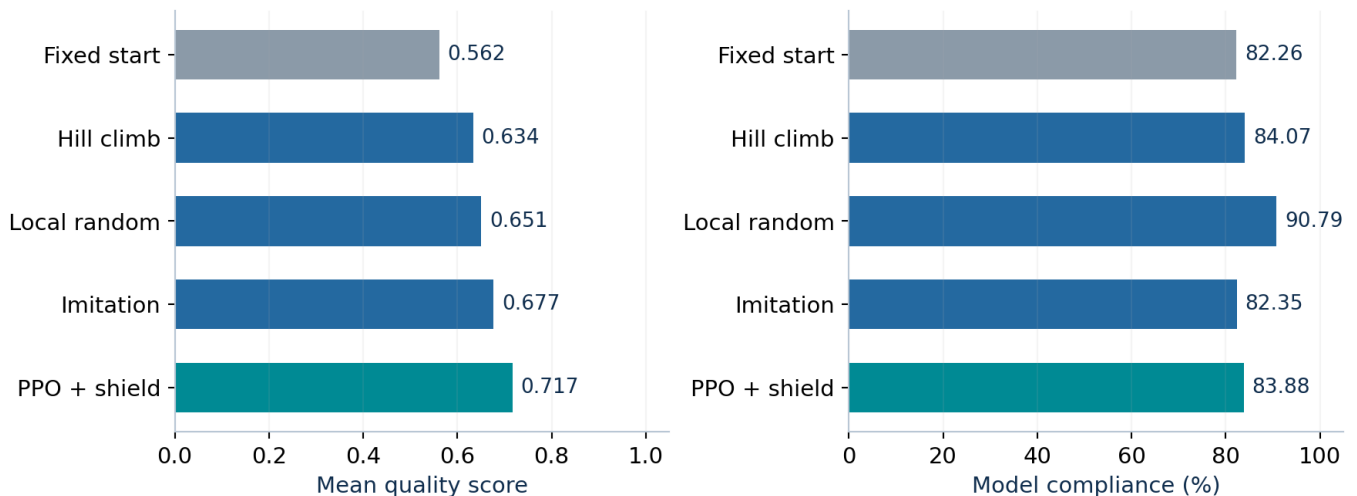


Figure 3. Exposed matched controls, eight-visit cap. PPO plus shield improves mean quality; local random exploration achieves higher compliance. All arms retain the fixed start. [E4]

The frozen FINAL evaluation contains 2,430 non-overlapping request/loss/PVT identities, not independent transistor geometries. Mean q rises from 0.5619 to 0.7169: gain 0.1550, paired 95% interval [0.1508, 0.1593]. Shielded compliance is 83.88%; raw policy compliance is only 64.44%. The shield is essential. [E3]

Interpretation. The study supports learned quality search within a characterized bank. It does not establish universal superiority, unrestricted MOS sizing or an on-chip controller with access to all verifier measurements.

The circuit and its physical bias

The physical export explains the CTLE core; the later receiver adds loaded voltage controls.

The source-degenerated CTLE

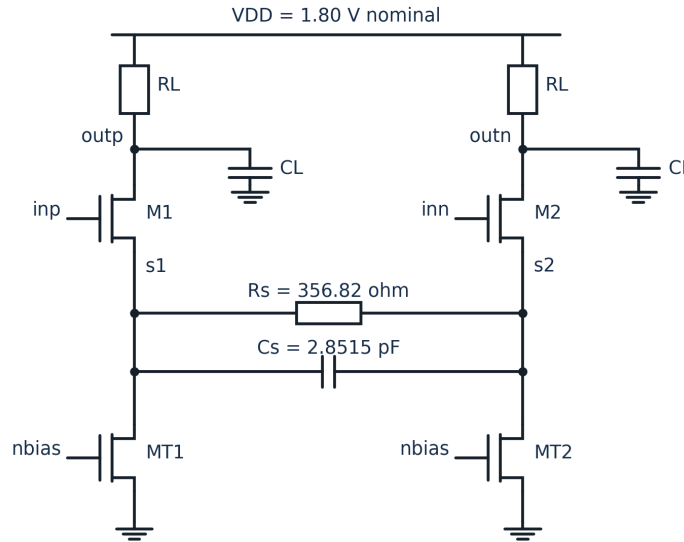


Figure 4. Earlier verified fixed physical export [E6]. Rs/Cs here are drawn fixed values; shared net labels connect the attenuator and bias. NMOS bulks connect to ground.

The input pair converts differential voltage into current; poly loads convert current back to voltage. At low frequency Rs provides local feedback and reduces gain. Cs bypasses that degeneration as frequency rises; output loading then limits the high-frequency response. Two tail devices preserve separate source nodes.

First-order reasoning. $f_z = 1/(2\pi R_s C_s)$, $k = 1 + (g_m + g_{mb}) R_s/2$, and A_{dc} is approximately $g_m R_L/k$. Rs is the full inter-source resistance. Body effect is included; finite output resistance and parasitics require the measured SPICE response. [E7]

Reference current made physical

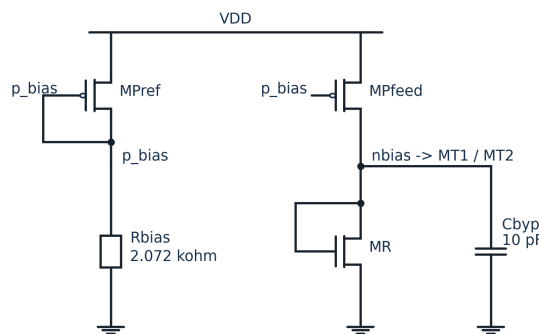


Figure 5. Physical PMOS reference/feed, poly bias resistor, diode-connected NMOS and MIM bypass [E6]. PMOS bulks connect to VDD. The bypass is a fixed capacitor.

VDD supplies a resistor-and-transistor reference; the diode-connected NMOS sets the shared tail-gate bias. A roughly 10 pF physical MIM bypass suppresses signal-frequency bias movement. This supply-dependent reference exposes mirror error, noise, power and area that ideal current sources conceal. It is not a precision bandgap.

Physical attenuation and peaking

Input excursion, relative peaking and absolute voltage gain answer different engineering questions.

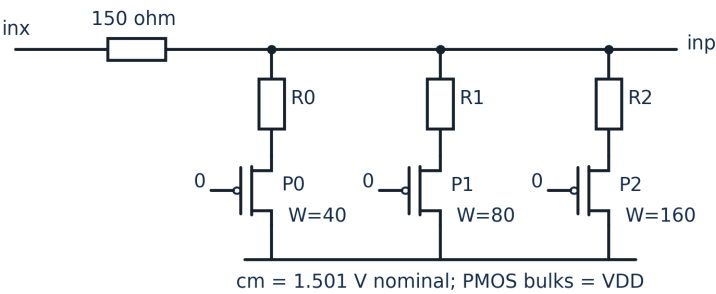


Figure 6. One leg of the earlier differential input attenuator [E6]; the other is identical. Three PMOS-switched shunts have real ON resistance and OFF parasitics.

The attenuator controls voltage presented to the active pair. This matters most on low-loss channels, where the minimum required CTLE boost can otherwise overdrive the stage. Peaking is a ratio to DC gain: about 9 dB of boost does not mean 9 dB of absolute gain. The input attenuator is not the source of CTLE peaking.

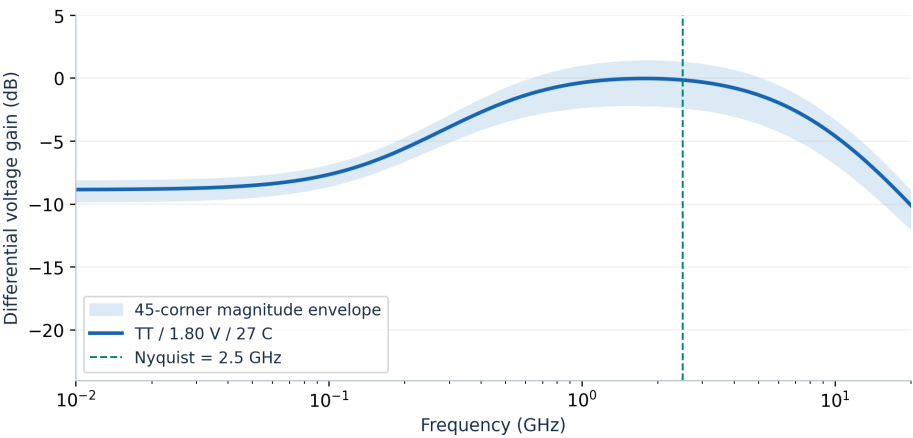


Figure 7. Absolute voltage gain and the 45-corner envelope of the earlier fixed CTLE [E6]. These curves do not establish analog PVT for the later integrated receiver.

From separate geometries to physical variable Rs/Cs

DESIGN STEP	FINDING / CONSEQUENCE
Original 512-setting bank	Attenuator switches are netlisted. Rs/Cs codes select separately drawn passives; a code table alone is not a physical switch matrix.
Series-switch studies	ON loss and OFF capacitance could not satisfy both registered limits at one width. Larger width did not solve the complete network.
Configurable receiver	A fixed poly Rs path parallels a resistor/NMOS path; two N500 varactors provide variable Cs. Filtered control feeds and bypass are physical.
Loaded calibration	At nominal VDD, R=1.26 V and C=0.333 V recover the 1.9 GHz identity after DFE loading. Signed Rs-switch Vds findings remain disclosed.

From transistor response to eye

A voltage-calibrated device-to-link bridge makes receiver quality affordable to inspect during search.

Saved transistor AC is fitted to a one-zero/two-pole CTLE model with a fit-validity gate, then combined with transmitter shaping and a constructed minimum-phase channel. The pulse response preserves voltage scale. Cursor samples separate desired signal from intersymbol interference without a full transistor transient for every candidate.

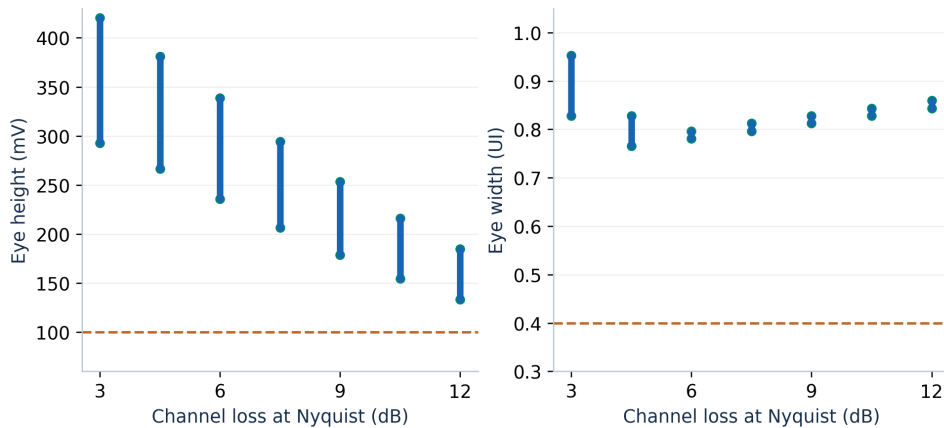


Figure 8. Earlier fixed physical CTLE: modeled ideal-DFE eye ranges across 45 corners at each of seven constructed losses, totaling 315 conditions. Heights and widths are cursor metrics. [E6]

What the 1-tap DFE contributes

For pulse cursors h_0 and h_1 , the normalized ideal tap is $b_1 = h_1/h_0$. Previous-bit feedback subtracts h_1 ; remaining pre- and post-cursors limit the eye. Ideal height is twice the difference between the main cursor and remaining absolute ISI sum, clipped at zero. Hardware additionally needs decision memory, timing and a physical summer.

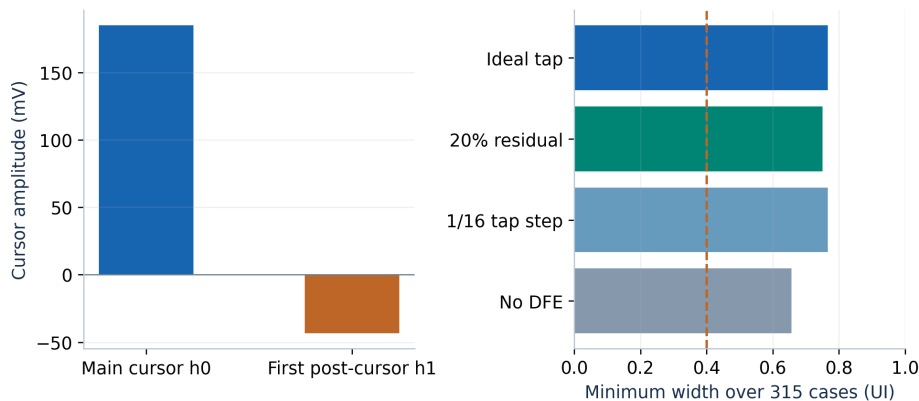


Figure 9. Saved behavioral comparison [E6]. This CTLE already meets eye targets without cancellation; ideal feedback chiefly increases width margin. Four model policies are not four transistor DAC implementations.

Measurement boundary. The model uses 0.8 V differential peak-to-peak TX, -3.5 dB de-emphasis, 64 samples/UI and 3-12 dB constructed losses. Eyes are noiseless ISI openings, not BER contours. Inferred phase, ideal decisions and omitted jitter limit the result; measured reflections and decision-error propagation are absent.

The integrated transistor receiver

A physical one-tap feedback path is a separate result from an ideal behavioral eye score.

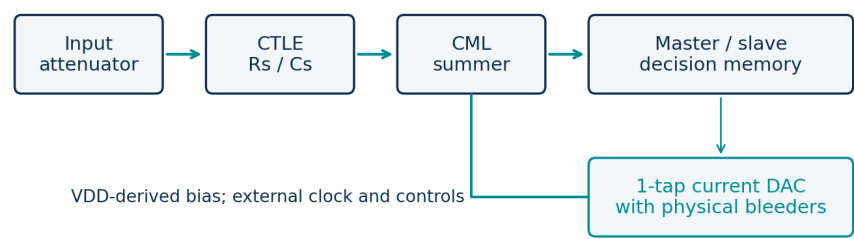


Figure 10. 73-device integrated checkpoint: configurable CTLE, CML summer, master/slave memory and current DAC. Physical bleeders prevent floating inactive DAC tails. [E1]

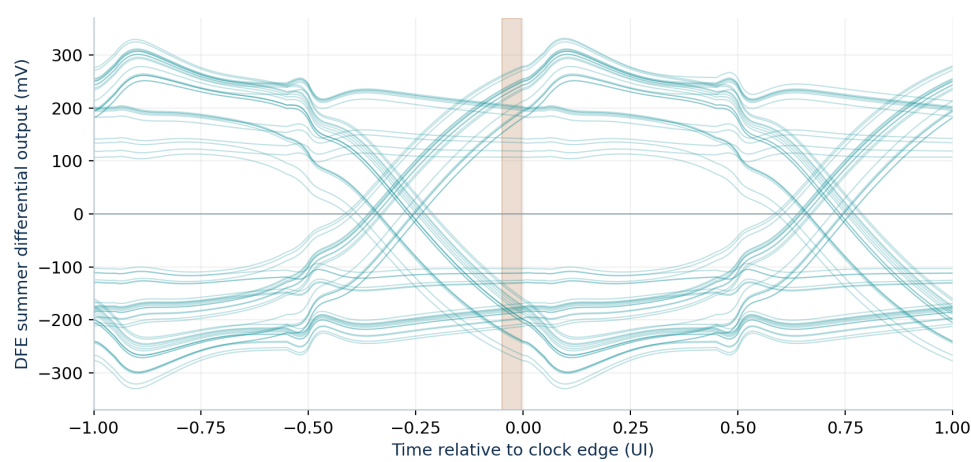


Figure 11. Saved transistor DFE-summer waveform overlay at nominal conditions [E1]. The finite, noiseless test scores 64 bits after settling on one constructed 7.5 dB channel.

What failed, and what changed

OBSERVATION	ENGINEERING RESPONSE / RETAINED LIMIT
Raw latch bits resolved; held ones failed	Storage and drive were tested separately. Wider CML memory later passed standalone PVT before feedback integration.
64/64 decoded; an inactive tail had reverse Vds	Added real discharge transistors. Logic did not waive the electrical gate; code zero became minimum current, not feedback off.
Standalone controls shifted near 1.75 GHz when loaded	Measured 65 loaded OP/AC points; selected .70/.185 VDD. Independent new analog/link checks followed; old passes were not transferred.
Runtime settling exceeded 500 ns	Longer standalone observation measured 850/690/650 ns. The original internal 500 ns criterion remains failed; DFE-active retuning is unverified.

Thought process. Every stronger test changed either the circuit, acceptance logic or claim. Frozen failed attempts remain available so progress can be checked rather than inferred from the final diagram.

Measured results and PVT scope

One fixed geometry, control fractions, tap code and clock phase at every declared link-PVT point.

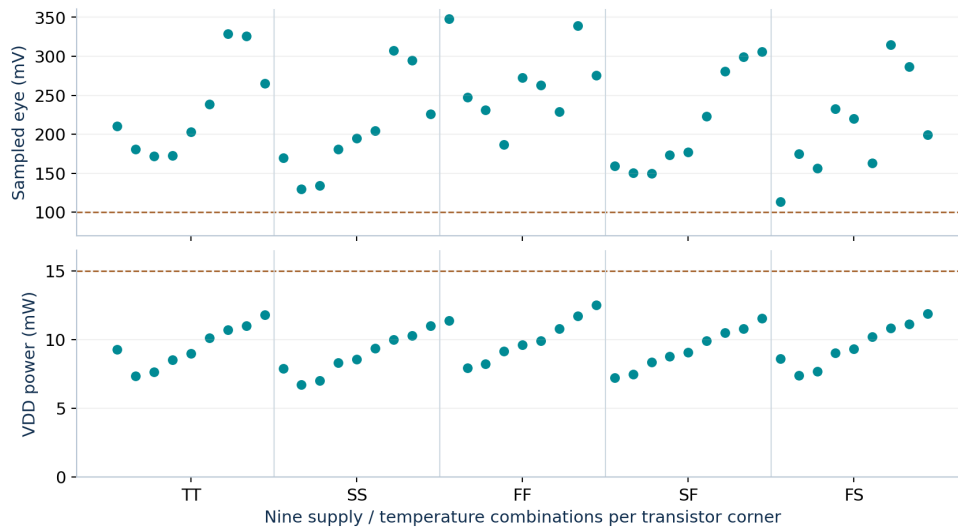


Figure 12. All 45 fixed-control link points, grouped by MOS process; each group includes three supplies and three temperatures. Dashed lines are the eye-height and VDD-power limits. [E2]

MEASUREMENT	RESULT	EXACT SCOPE
Nominal peaking / peak	8.744-8.747 dB 1.903-1.906 GHz	Loaded held-clock states; .5 dB / 100 MHz internal request tolerances.
Input-referred noise	0.65404-0.65405 mVrms	10 MHz-5 GHz; two nominal negative-branch held-clock states, not periodic noise.
Clocked HD3	-54.9201 to -54.9137 dBc	100 MHz, 100 mV differential peak; four time-step/tolerance settings, CTLE output.
Link PVT	45/45; 64/64 bits each	TT/SS/FF/SF/FS; 1.71/1.80/1.89 V; 0/27/125 C; constructed 7.5 dB channel.
Minimum eye / width	113.244 mV / 0.635 UI	Positive opening width; width above 100 mV is 0.560 UI. Distinct definitions.
Maximum VDD / geometry	12.5241 mW 0.014715766 mm ²	External clock/control/common-mode sources excluded; geometry subtotal is not routed area.

Scope of the result. These are nominal analog checks plus sampled link PVT, not simultaneous all-specification PVT. Signed bilateral Rs-switch Vds findings persist at all 45 points. Omitted poly-resistor voltage dependence, passive corners, mismatch, periodic noise, full clock/control power and layout remain open.

Deadline pilot. A separately registered two-corner analog pilot reproduced the nominal checks; FS / 1.71 V / 125 C failed the required latch initialization. It adds no valid analog-PVT coverage. [E10]

An inspectable automation receipt

A saved production run exposes the policy, verifier and classical recovery as separate contributors.

The receipt reconstructs the existing 9 dB / 1.9 GHz adaptive-bank run; it does not rerun optimization. Its representative exported CTLE is setting 425. The 315 accepted conditions form an adaptive code map, distinct from the fixed 3 dB physical demo and the separate transistor receiver. [E8]

RECORDED STAGE	MEASURED ACCOUNTING
Policy and verifier	2,406 visits/checks, including 315 fixed starts and repeats. 81 outcomes retained by the RL-visit shield.
Classical selection	185 target-centering outcomes + 49 safety recoveries = 234 bank-selected outcomes; 119,808 table rows checked.
Search / export cost	Zero new search/verification SPICE calls; one representative export call. Recorded design-plus-deck timer 5.894 s; export component 0.225 s. Drawing/writing excluded.

An actual safety-recovery trace. FS / .95 VDD / 125 C at 3 dB channel loss: 360 > 296 > 232 > 168 > 104 > 168 > 232. No compliant visited candidate was retained; the bank supplied setting 496. This final code is not an actor visit.

Search effort is not elapsed design time

COMPARISON	RESULT / INTERPRETATION
Policy vs cached exhaustive oracle	5.579 versus 512 visits on average; 91.8x fewer candidate visits. This is not an end-to-end SPICE or wall-clock speedup. [E5]
Near-optimality	Mean oracle regret 0.28310; 20.91% within 0.05 of oracle quality. Registered near-optimality gate fails.
PPO attribution	At cap eight, PPO quality exceeds imitation by 0.039957, but local random compliance is higher. The all-classical-controls no-loss condition is not met. [E4]

The historical bank, midpoint and five PPO stages sum to 6.711 recorded hours. This excludes imitation, teacher construction, earlier sizing/training and other audits; original bank retries were not fully billed. It is not total project duration. No matched end-to-end MOS/R/C/L sweep establishes a final-physical speedup. [E4]

Competitive contribution. Bounded learned search, explicit recovery and traceable output reduce opaque manual decisions. The evidence supports that integrated workflow without treating cached visits as seconds saved.

The designer workflow

The desktop workbench organizes inspection around one selected completed result.

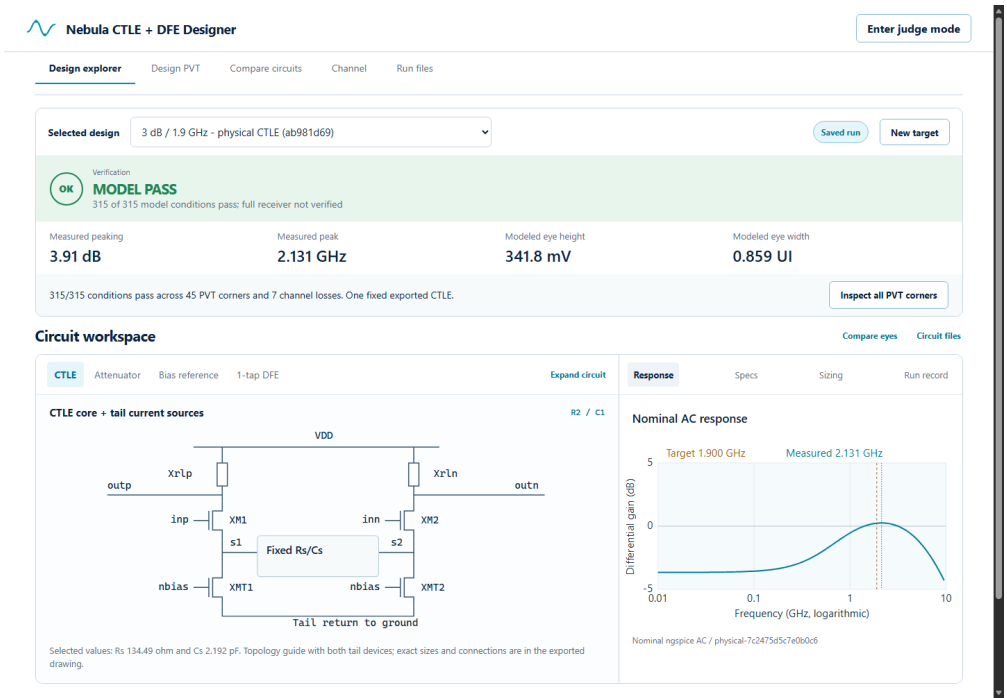


Figure 13. Design Explorer from the local application. A shared circuit canvas and adjacent inspector connect the selected circuit to response, specifications, sizing and its run record. [E9]

DEMO STEP	WHAT THE REVIEWER CAN INSPECT
Design Explorer	Open the saved 3 dB / 1.9 GHz physical result: measured 3.910115 dB / 2.131144 GHz; modeled eye 341.791199 mV / 0.859375 UI. Start with CTLE and tails.
Design PVT	315/315 model conditions = 45 PVT corners x seven losses. This physical export uses one fixed setting; each displayed heatmap is one 45-point loss slice.
Compare circuits	CTLE-only, CTLE + ideal DFE and separate margin-envelope views. Waveforms come from saved AC; recorded dimensions remain cursor-based metrics.
Separate transistor checkpoint	Expand the labeled 9 dB / 1.9 GHz evidence: actual DFE, nominal analog results and 45-point finite-pattern transistor link PVT.
Channel / Run files	Profile an uploaded Touchstone file without inheriting bank verification. Inspect exact exported decks, specifications, drawings and evidence.

The saved 3 dB request uses production tolerances of 1.5 dB and 0.3 octave; its response is accepted within those tolerances, not exactly equal to the target. Editing the next request never relabels the completed result. A replayed demonstration is identified as saved evidence, not newly finished live optimization.

Deliverable closure. Framework, circuit outputs, measurements and evidence are ready to inspect. The demo route above connects the report evidence to the actual workflow. Complete tuning coverage, extracted silicon area, all-specification PVT and provider-backed LLM fine-tuning remain further work.

Conclusions and evidence guide

A working, inspectable design framework with an independently measured physical receiver checkpoint.

Nebula addresses the brief with target input, learned proposals, deterministic acceptance, open-source SPICE integration and circuit/specification output. The hardware extension demonstrates calibrated variable Rs/Cs and a transistor one-tap DFE at one loaded target. It strengthens implementation evidence while preserving the distinction between learned bank search and engineer-developed circuitry.

Next engineering steps. Close analog PVT and loaded target coverage; resolve model-domain and passive-model limits; add DFE-active retuning, clock/control generation and extracted layout. Compare future automation against strong classical methods with matched complete timing and simulator budgets.

Source-to-claim guide

KEY / PAGES	SOURCE ARTIFACT (RELATIVE TO REPOSITORY)
E1 / 1,2,5-9	nebula/DFE_CALIBRATED_RESULTS.md; nebula/product_audits/entry143_dfe_calibrated_verification_20260910/ - nominal checks, exact transistor deck and waveform.
E2 / 1,2,8,9	nebula/product_audits/entry144_dfe_calibrated_pvt_20260910/ - fixed-control link PVT, all 45 decks, summary and lossless raw archive.
E3 / 3,4	nebula/experiments/shielded_policy_final_results.json; nebula/rl/hybrid_designer.py - frozen final study and actual production logic.
E4 / 4,10	nebula/POST_REVIEW_RESULTS.md; nebula/product_audits/entry113_attribution_20260907/ - matched controls, failures and cost ledger.
E5 / 10	nebula/product_audits/entry115_exhaustive_benchmark_20260908/ - cached oracle and failed near-optimality gate.
E6 / 3,5-7	nebula/product_demo/physical_bias_9db_1p9ghz_20260906/; nebula/product_audits/entry115_physical_recovery_20260908/ - fixed CTLE and recovered 3/6 dB targets.
E7 / 1,2,5	CLAUDEwa.md - deliverables, S1-S9, circuit conventions. Pre-existing SerDes infrastructure is declared; reported results are Nebula NRZ only.
E8 / 10	nebula/product_demo/rl_hybrid_9db_1p9ghz/design.json; nebula/AUTOMATION_RECEIPT_20260914.md - actual policy traces, selection and partial timing.
E9 / 11	nebula/web/; nebula/report/final_submission_assets/desktop.png - five-tab implementation and captured workbench.
E10 / 9	nebula/product_audits/submission_analog_pvt_20260914/second_attempt/summary.json - registered two-corner analog pilot; stress initialization failed.

Methods and tools. [R1] SkyWater SKY130 official documentation. [R2] ngspice manual and documentation. [R3] Schulman et al., *Proximal Policy Optimization Algorithms*, 2017, arXiv:1707.06347. Measurements are E1-E10, not values borrowed from these references.

Build: `py -3.13 -m nebula.report.final_submission`; check: `py -3.13 -m nebula.report.check_final_submission`. Companion sources/review JSON files record hashes and page checks. Three earlier PDFs are preserved. Local evidence validates; nine legacy JSON newline mismatches remain a documented clean-clone rebuild limitation (HANDOFF G197).